of that concept–format. We then form one vector per format by summing these mean activations over the top-$K$ heads selected for $\mathcal{CV}$ or $\mathcal{FV}$ (as in Eq. 5, but using per-format means in place of per-prompt activations). This yields one ID vector (`OE-ENG`) and two OOD vectors (`OE-FR`, `MC`) per concept.

**AmbiguousICL Task.** We evaluate on AmbiguousICL tasks (Figure 6): each prompt interleaves two concepts (3 then 2 exemplars) followed by a query. The second concept is always English→French translation. Unsteered models tend to continue with the second concept; we aim to steer toward the first. Note that steering prompts are distinct from the extraction prompts used to construct the vectors. This setup is diagnostic: it tests whether representations encode abstract relational structure independent of extraction prompts' surface format. To perform well in this setup requires consistency between ID and OOD performance.

**Steering with $\mathcal{CV}$s and $\mathcal{FV}$s.** We add a vector $\mathbf{v}$ to the last-token residual stream at a chosen layer:

$$\mathbf{h}_\ell \leftarrow \mathbf{h}_\ell + \alpha \mathbf{v} \tag{6}$$

We measure effectiveness as $\Delta P = P_{\text{after}}(y) - P_{\text{before}}(y)$, averaged over 100 prompts per concept (see Figure 22 for Top-1 accuracy). We sweep $\alpha$ and $K$ and report the best per model (Appendix F).

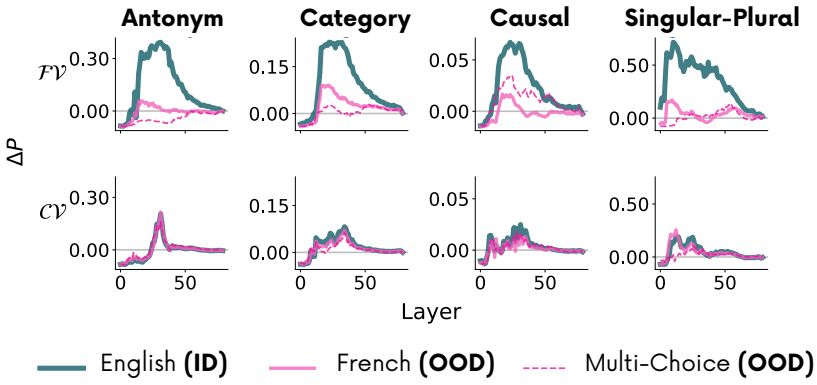


Figure 7: *Steering effect across layers*. We inject $\mathcal{CV}$s and $\mathcal{FV}$s into Llama-3.1-70B and plot the change in target-token probability ($\Delta P$) for four representative concepts (columns). Curves compare ID extraction format with OOD formats relative to the AmbiguousICL task (Figure 6). Higher $\Delta P$ means the model assigns more probability to the expected token than the unsteered model. **Takeaways**: (1) $\mathcal{FV}$s typically achieve larger ID gains but often drop OOD. (2) $\mathcal{CV}$s yield smaller gains yet show more stable OOD behavior across formats. See Figure 16 for other concepts/models.

## 3.2 Steering Results

### 3.2.1 Function Vectors Outperform Concept Vectors in Distribution

Extracted from `OE-ENG` (ID setting), $\mathcal{FV}$s yield the largest gains on ambiguous prompts (Figure 7). $\mathcal{CV}$s also help but with smaller $\Delta P$ and minimal zeroshot effect (Figure 17). At the token level both vectors lift plausible English antonyms in the ID case (Table 2).

### 3.2.2 Concept Vectors Are More Stable Out of Distribution

**Performance gains ($\Delta P$).** Out of distribution (extracting vectors from `OE-FR` or `MC`), $\mathcal{CV}$s more often maintain positive effects across formats, whereas $\mathcal{FV}$s frequently degrade—especially for `MC`—and only occasionally stay consistent for specific concepts/models (Figs. 7, 16). $\mathcal{CV}$s raise the probability of the correct English answer across formats, and their top-$\Delta$ tokens remain concept-aligned (Table 2). Crucially, the key finding is not absolute performance but consistency: $\mathcal{CV}$s increase the probability of producing similar concept-aligned tokens regardless of extraction format.

**Distributional consistency (KL).** To quantify consistency across formats independent of absolute gains, we compare the model's next-token distributions after steering with ID and OOD vectors. For

**Query:** `salty →`

| | + Antonym | Top Δ Tokens |
|---|---|---|
| $\mathcal{FV}$ | OE-ENG | _sweet (+56%), _fresh (+16%), _bland (+6%), _taste (+3%), _uns (+2%) |
| | OE-FR | _su (+31%), _dou (+27%), _frais (+5%), _fade (+5%), _ins (+3%) |
| | MC | _( (+53%), _A (+1%), _\n (+1%), _space (+0%), _) (+0%) |
| $\mathcal{CV}$ | OE-ENG | _sweet (+49%), _fresh (+8%), _bland (+3%), _taste (+3%), _uns (+3%) |
| | OE-FR | _sweet (+54%), _fresh (+9%), _bland (+3%), _uns (+3%), _taste (+2%) |
| | MC | _sweet (+35%), _fresh (+12%), _bland (+4%), _uns (+3%), _taste (+3%) |

Table 2: *Token-level steering effects*. Top tokens with largest probability gains when injecting $\mathcal{CV}$s or $\mathcal{FV}$s into Llama-3.1-70B on the AmbiguousICL prompt (query shown above). Results shown at the layer with the strongest in-distribution effect per vector. Without intervention, the model predicts French translation _sa (from *salé*) with 49%; antonym _sweet has 2%. English antonyms in red, French in blue, and the opening bracket (MC token) in green.

each concept and vector type, we select the top 5 layers that achieve the highest ID $\Delta P$. At each selected layer we compute KL divergence

$$D_{KL}\left[p(\mathbf{x} \mid \mathbf{v}_{\text{OOD}}) \,\|\, p(\mathbf{x} \mid \mathbf{v}_{\text{ID}})\right]$$

between the post-intervention distributions at the query token, where lower values indicate more similar effects of ID and OOD vectors. We average this KL divergence over prompts and selected layers to obtain one score per concept, and then summarize per model (Figure 8). Across models, $\mathcal{CV}$s yield lower KL than $\mathcal{FV}$s. The CV–FV KL gap is larger for MC than for OE-FR.

### 3.2.3 Function Vectors Mix Concept with Input Format

Out of distribution, $\mathcal{FV}$s reflect both prompt format and concept. When vectors are extracted from OE-FR, they push the model toward the French translation of the concept (e.g., French antonyms), and when extracted from MC, they increase the probability of format tokens such as the opening bracket (Table 2). We quantify the language effect by measuring $\Delta P$ for the French translation across concepts (Figure 13). In the larger models, $\mathcal{FV}$s substantially increase the probability of the French token, whereas $\mathcal{CV}$s remain near zero; in smaller models the effect is negligible. Notably, $\mathcal{FV}$s extracted from open-ended Spanish prompts induce almost the same bias toward the French translation as $\mathcal{FV}$s extracted from French prompts (Figure 14), even though the AmbiguousICL alternatives are French only. This pattern suggests that $\mathcal{FV}$s capture a generic translation/foreign-language signal tied to the extraction format rather than language-specific content. Combined with the MC bracket effect (Figure 15), these findings indicate that $\mathcal{FV}$s mix concept with surface format, while $\mathcal{CV}$s are comparatively format-invariant.

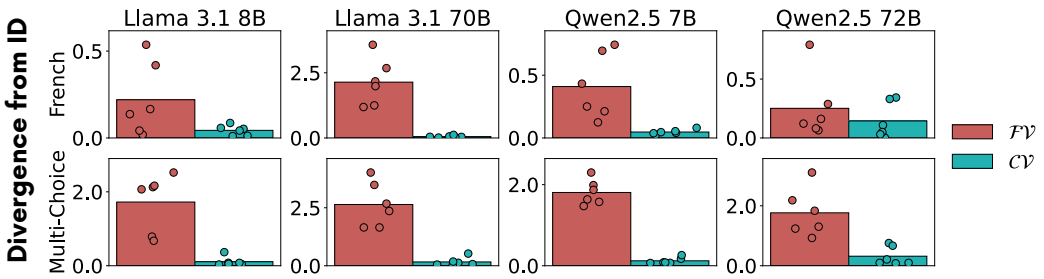


Figure 8: *Consistency of steering effects*. KL divergence between the probability distributions after the models were steered with an ID (OE-ENG) and OOD vectors (OE-FR [top], MC [bottom]). Lower values indicate more similar effects of ID and OOD vectors. **Takeaway**: $\mathcal{CV}$s steer the models more consistently than $\mathcal{FV}$s.

## 4 Related Work

**Attention Head Categorization.** Recent work has made significant progress in characterizing specialized attention heads that process in-context learning (ICL) tasks. For instance, Olsson et al. (2022) identified induction-heads, which Yin & Steinhardt (2025) found can develop into $\mathcal{FV}$-heads during training. Other specialized head types include semantic-induction heads (Ren et al., 2024), symbol-abstraction heads (Yang et al., 2025), and various others (Zheng et al., 2024). Our work extends this line of research by identifying $\mathcal{CV}$ heads, attention heads that invariantly represent concepts in ICL tasks at high levels of abstraction.

**Linear Representation of Concepts.** A substantial body of research has established that concepts are represented linearly in LLMs' representational space (Mikolov et al., 2013; Arora et al., 2016; Elhage et al., 2022). This phenomenon, often termed the "Linear Representation Hypothesis" (Park et al., 2024), has been extensively studied across various tasks and domains. Hernandez et al. (2024) demonstrated that relational concepts—similar to those we study in this paper—can be decoded from LLM activations using linear approximation. Subsequent work by Merullo et al. (2025) revealed that the success of such decoding depends on the frequency of concepts in the pretraining corpora, which may explain why some concepts are represented more consistently than others in our study. Our findings contribute to this literature in two ways: (1) providing further support for the Linear Representation Hypothesis, and (2) extending previous work on relational concept representations by localizing specific attention heads that carry such representations and demonstrating their invariance to input formats.

**Symbolic-like reasoning in LLMs.** Recent work has demonstrated that LLMs can exhibit symbol-like representational properties even without explicit symbolic architecture (Feng & Steinhardt, 2024; Yang et al., 2025; Griffiths et al., 2025). Yang et al. (2025) define symbolic processing as requiring two key properties: (1) invariance to content variations, and (2) indirection through pointers rather than direct content storage. Our $\mathcal{CV}$s exhibit both properties: they are invariant to input format changes and function as pointers to content stored elsewhere, unlike $\mathcal{FV}$s which directly store content (§3.2.3).

## 5 Discussion

Our results separate two representational roles in LLMs: components that *cause* strong ICL performance and components that *encode* abstract concept structure. Function Vectors ($\mathcal{FV}$s) occupy the first role, steering models effectively when extraction and application formats match, but deteriorating out of distribution (formats/languages). Conversely, Concept Vectors ($\mathcal{CV}$s) built from RSA-selected heads encode concepts at a *higher level of abstraction* and generalize more robustly across languages and question types, albeit with smaller causal effects. This supports a view that invariance and causality are mediated by largely distinct mechanisms in similar layers.

**Layers of abstraction.** We define abstraction as encoding relational structure (e.g., "antonym") while discarding surface details (e.g., "English, multiple-choice"). Our results suggest $\mathcal{FV}$s do capture abstract task information: they reliably encode concepts within a format (Figure 19) and are causally effective even across formats (Appendix O). However, their orthogonality across formats and retention of surface signals (e.g., MC brackets) reveal that they conflate this abstract content with surface form. In contrast, $\mathcal{CV}$s cluster by concept regardless of format. Thus, $\mathcal{FV}$s operate at a lower level of abstraction ("antonym in MC format"), while $\mathcal{CV}$s operate at a higher level ("antonym"), independent of surface form.

**Relation to Function Vectors.** Prior work shows that $\mathcal{FV}$s compactly mediate ICL and can transfer across contexts (Todd et al., 2024). We refine this: $\mathcal{FV}$ portability is strong within families of prompts, but is not fully invariant to surface format. Same-concept $\mathcal{FV}$s extracted from different formats are nearly orthogonal and can carry language/format signals (e.g., French subword or multiple-choice bracket tokens), while $\mathcal{CV}$s track concept across formats with less surface content. This distinction can be framed as *equivariance* vs. *invariance*: $\mathcal{FV}$s adapt to extraction format (e.g., producing French antonyms from French prompts, MC formatting tokens from MC prompts), whereas $\mathcal{CV}$s steer toward similar outputs regardless of format. Finally, we do not propose $\mathcal{CV}$s as competitors to $\mathcal{FV}$s, but rather highlight a mechanistic dissociation: $\mathcal{FV}$s drive behavior (causality) while $\mathcal{CV}$s represent abstract structure (invariance).